Assignment

Q1 Make a redux app to make a counter which will start from 0 and 2 buttons with increment and decrement options and count will increase and decrease on the basis of which button.

Ans Basic objectives :-

(1) Create a Redux - based counter app:
- Initial app : 0 (value)
- Two buttons:
  - Increment (+)
  - decrement (-)
- Count updates based upon button clicks.

Concepts used :-

- Redux Toolkit (@reduxjs/toolkit)
- React - Redux (useSelector, useDispatch)
- Centralized state management

Steps to make the app :-

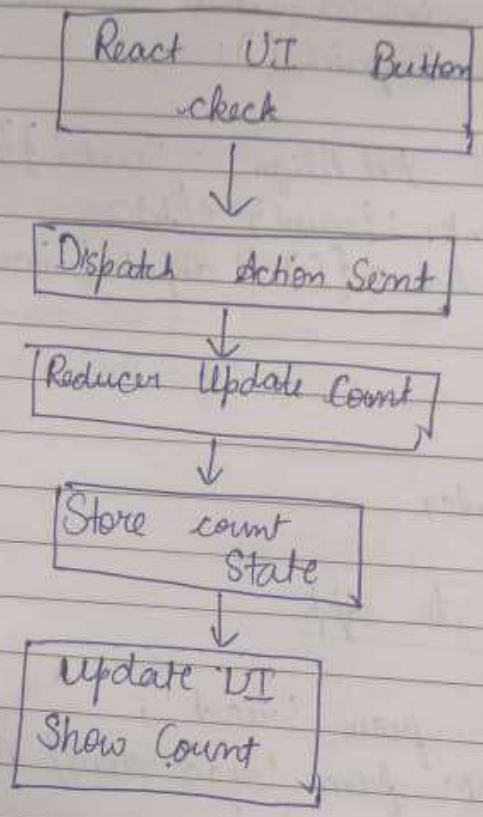
(1) Install Dependencies

```
npm install @reduxjs/toolkit react-redux
```

(2) Create Redux Store

```
import { configureStore } from "@reduxjs/toolkit";
import counterReducer from "../features/counter/counterSlice";
Export const store = configureStore({
```

Diagram :-


React UI Button
check
Dispatch Action Sent
Reducer Update Count
Store count
State
Update UI
Show Count

```
const Counter = () => {
  const count = useSelector((state) => state.counter.value);
  count dispatch = useDispatch();
  return (
    <div style {{ text Align : 'center' }} >
      <h1> Counter : {count} </h1>
      <button onclick = {() => dispatch(increment())} > </btn>
    </div>
  );
};
export default counter
```

5. Connect Redux to App

```
import React from 'react';
Import React DOM from 'react-dom/ client';
import App from './App';
import { Provider } from 'react-redux';
React DOM.create Root(document.get ElementById ('root')).render(
    <Provider store = {store}>
      <App>
);
```

6. Use counter in App

```
import counter from './location';
function App () {
  return (
    <div>
      <counter />
    </div>
  );
}
```

```
    reducer : {
        counter : counterReducer,
    },
});
```

③ Create Counter Slice

```
import { create Slice } from '@ reduxjs/toolkit';
const initial State = {
    value : 0;
};
const counterSlice = Create Slice ({
    name : 'counter',
    initial State,
    reducer : {
        increment : (state) => {
            state.value += 1;
        },
        decrement : (state) => {
            state.value -= 1;
        },
    },
});
export const { increment, decrement } = counterSlice.actions;
export default counter Slice.reducer;
```

④ Create counter Component

```
import React from 'react';
import { useSelector, Use Dispatch } from 'react-redux';
import { increment, decrement } from './counterSlice';
```